

Simulation Modes and Complexity Toggles

Purpose

Not every run needs every feature. This document defines the configuration switches that control the simulation's scope and complexity. Simpler runs are faster, easier to debug, and better for isolating specific effects. Complex runs are richer and more realistic.

All toggles are set in config.yaml before the run starts and are fixed for the duration of the run.

Simulation Mode: Public vs. Private

The Choice

At run start, the user chooses how firms begin:

```
simulation:
  mode: "private_start"      # or "public_start"
  allow_ipo: true           # if private_start, can firms IPO during the run?
```

Mode: public_start (Default)

Firms begin with an IPO in Q1. The Equity Market agent acts as the public stock market from the start.

Q1 sequence:

- Firm requests capital raise
- Investment Bank structures the IPO (share count, price range, prospectus)
- Equity Market sets the market-clearing price and subscribes to shares
- Capital is booked; firm has public equity and a stock price
- Firm begins operating

Ongoing: Equity Market sets a stock price every quarter. Secondary offerings go through IBank structuring then Equity Market subscription. Shares are freely tradeable (modeled as continuous market pricing by the Equity Market agent).

Mode: private_start

Firms begin as private companies funded by the Equity Market agent acting as a PE/VC consortium.

Q1 sequence:

- Firm pitches for funding (capital need, business plan, equity offered)
- Investment Bank advises on deal structure (valuation, terms, governance)
- Equity Market (in PE mode) decides: fund or pass? At what valuation?
 - Sets a "post-money valuation" and takes an equity stake
 - May impose governance terms: board seat, anti-dilution, liquidation preference
- Capital is booked; firm has private equity at the agreed valuation
- Firm begins operating

Ongoing: No public stock price. Equity Market sets a "last-round valuation" each quarter (a private mark-to-market, like a VC portfolio valuation). Firms can raise additional private rounds (same process: firm requests -> IBank structures ->

Equity Market decides).

Key differences from public mode:

Aspect	Private	Public
Equity pricing	Last-round valuation (PE markup)	Market price (public trading)
Pricing frequency	Each funding round + quarterly mar	Every quarter
Information available	Extensive due diligence (more than	Published financials only
Liquidity	Zero (no secondary market)	Full (shares trade freely)
Governance	Board seats, veto rights, preferen	Shareholder voting only
Dilution mechanics	Down rounds, anti-dilution ratchet	Market-price issuance
Cost of capital	Higher (illiquidity premium)	Lower (liquid market)

IPO Transition (if allow_ipo: true)

A private firm can choose to IPO at any point. The IPO decision is part of the firm's quarterly decision:

```
{
  "ipo_request": {
    "go_public": true,
    "desired_raise": 500000000,
    "shares_to_offer_pct": 0.25,
    "reasoning": "We need growth capital and our PE investors want liquidity."
  }
}
```

IPO process:

- IBank evaluates and structures the offering (price range, share count, roadshow)
- Equity Market (switching from PE to public mode for this firm) sets the IPO price
- If Equity Market subscribes: IPO succeeds, firm becomes public
- If Equity Market declines (price too high, market conditions bad): IPO fails,

firm remains private, can try again next quarter

- Existing PE investors' shares convert to public shares at IPO price

After IPO: The firm is treated as public for all future quarters. The Equity Market agent prices it as a public stock. PE-era governance terms expire.

When Would a Firm IPO?

The firm agent (and the environment narrative) should consider:

- Cash need: Private rounds become expensive (high dilution) for mature companies
- PE investor pressure: After 8-12 quarters, PE investors want exits
- Market conditions: IPO windows open and close with market sentiment
- Competitive positioning: Being public signals maturity and credibility
- Regulatory: Full ALT approval may require public disclosure levels

Complexity Toggles

Toggle Table

```
complexity:
```

```
# Core toggles (recommended for all runs)
entry_exit: true           # firm bankruptcy, entry, death-spiral prevention
financial_institutions: true # banks and equity market; if false, fixed financing

# Feature toggles (activate for richer runs)
ma_enabled: false          # M&A proposals, goodwill, impairment
leasing_enabled: false      # lease vs. build, ROU assets, lease liabilities
stock_comp_enabled: false   # option/RSU grants, diluted shares, non-cash expense
workforce_detail: false     # hiring/layoff decisions, workforce effects
working_capital_decisions: false # collection, inventory, payment terms
provisions_enabled: false    # litigation reserves, warranty provisions
voluntary_liquidation: false # firms can choose to wind down

# Product/market toggles
multi_product: false        # multiple product lines per firm (NOT v1)
regions: false              # geographic market segmentation (NOT v1)

# Accounting toggles (orthogonal to above)
nol_carryforward: true      # tax loss carryforward tracking
fair_value_regime: false     # PPE marked to fair value (vs. historical cost)
rd_capitalization: false     # capitalize portion of R&D (vs. expense all)
```

What Each Toggle Controls

entry_exit: true (default ON)

ON: Firms can default. Vacant slots may receive new entrants (environment decides). Death-spiral tracker active.

OFF: Firms never default. If cash goes negative, orchestrator injects emergency equity (logged as "government bailout") at dilutive terms. Useful for studying firm behavior without bankruptcy risk.

financial_institutions: true (default ON)

ON: 4 financial agents (Equity Market, IBank, Commercial Bank, Credit Fund) operate independently with LLM-driven decisions.

OFF: Financing is on autopilot. Orchestrator provides:

- Fixed revolver at $r_f + 300\text{bps}$, commitment = 1x quarterly revenue
- Fixed term debt at $r_f + 500\text{bps}$, max = 0.5x total assets
- Equity price = book value (no market pricing)
- No equity subscription decisions (firms issue freely)

Useful for studying firm strategy in isolation from capital market effects.

ma_enabled: false (default OFF)

ON: Firms can propose acquisitions. Full M&A process: financing, goodwill, impairment testing, integration effects. IBank advises on deals. Equity Market prices post-merger equity. Creates vacant slots for potential entry.

OFF: No M&A. The `ma_proposal` field is removed from firm decisions. No goodwill on balance sheets. Simpler accounting.

leasing_enabled: false (default OFF)

ON: Firms choose lease vs. build for capacity. Lease decisions added to firm JSON. ROU assets and lease liabilities on balance sheet. Lease expense flows through P&L.

OFF: All capacity expansion is via capex (purchase). No lease-related balance sheet items. rouq, leaseq, xlrq columns are zero.

stock_comp_enabled: false (default OFF)

ON: Firms grant options and RSUs. Non-cash SGA expense. Diluted shares tracked. EPS computed on diluted basis.

OFF: All compensation is cash. Basic shares = diluted shares. stkcpq column is zero.

workforce_detail: false (default OFF)

ON: Firms make hiring/layoff decisions by category. Workforce count affects R&D speed (scientists), manufacturing quality (ops staff), and commercial effectiveness (sales). Restructuring charges on layoffs.

OFF: Workforce is modeled implicitly. Headcount scales automatically with revenue/R&D/SGA spending. No explicit workforce decisions. empq is computed from a formula, not a decision.

working_capital_decisions: false (default OFF)

ON: Firms choose collection aggressiveness, inventory targets, supplier payment terms. Working capital becomes a strategic lever.

OFF: Working capital uses defaults: AR = 15% of revenue, AP = 15% of COGS, accrued = 10% of (SGA + R&D). Firms cannot influence these ratios.

provisions_enabled: false (default OFF)

ON: Firms decide litigation reserve additions, warranty provisions. Provisions appear as long-term liabilities. Settlements reduce provisions.

OFF: Litigation costs are rolled into SGA automatically (2-4% of revenue). No explicit provision decisions or balance sheet items.

Recommended Configurations

Minimal (fastest, for testing and debugging)

```
simulation:
  mode: "public_start"
  n_firms_initial: 3
  n_quarters: 20
complexity:
  entry_exit: true
  financial_institutions: false    # autopilot financing
  ma_enabled: false
  leasing_enabled: false
  stock_comp_enabled: false
  workforce_detail: false
  working_capital_decisions: false
  provisions_enabled: false
```

3 firms, 20 quarters, no banks, no M&A, no leasing, no stock comp. Focus on product market competition and R&D race.

Standard (balanced realism and speed)

```
simulation:
```

```

mode: "public_start"
n_firms_initial: 5
n_quarters: 40
complexity:
  entry_exit: true
  financial_institutions: true
  ma_enabled: false
  leasing_enabled: false
  stock_comp_enabled: false
  workforce_detail: false
  working_capital_decisions: false
  provisions_enabled: false

```

5 firms, 40 quarters, full financial agents, but no optional complexity. This is the recommended starting point for research runs.

Full (maximum realism)

```

simulation:
  mode: "private_start"
  allow_ipo: true
  n_firms_initial: 5
  n_quarters: 80
complexity:
  entry_exit: true
  financial_institutions: true
  ma_enabled: true
  leasing_enabled: true
  stock_comp_enabled: true
  workforce_detail: true
  working_capital_decisions: true
  provisions_enabled: true

```

Private start with IPO option, all features on, 80 quarters. Richest simulation but slowest (most LLM calls, most complexity).

Research: Regime Comparison

```

# Run A: Baseline
simulation:
  mode: "public_start"
  seed: 42
information_regime: "baseline"
measurement_regime: "baseline_gaap"

# Run B: Same seed, different measurement
simulation:
  mode: "public_start"
  seed: 42
information_regime: "baseline"
measurement_regime: "rd_capitalization"

```

Same firms, same shocks, different accounting rules. Compare outcomes.

How Toggles Affect Agent Prompts

When a toggle is OFF, the corresponding section is omitted from prompts entirely. Agents are not told "M&A is disabled" -- they simply never see the M&A decision field or any M&A-related context. This prevents agents from reasoning about features that don't exist in the current run.

When a toggle is ON, the corresponding decision fields, context sections, and world doc references are included in prompts.

The prompt builder checks toggles:

```
def build_firm_prompt(firm_state, config):
    prompt = base_prompt(firm_state)

    if config.complexity.ma_enabled:
        prompt += ma_decision_section(firm_state)
    if config.complexity.leasing_enabled:
        prompt += lease_decision_section(firm_state)
    if config.complexity.stock_comp_enabled:
        prompt += stock_comp_section(firm_state)
    if config.complexity.workforce_detail:
        prompt += workforce_section(firm_state)
    # etc.

    return prompt
```

How Toggles Affect Compustat

All Compustat columns are ALWAYS present in the panel (for schema consistency). When a toggle is OFF, the corresponding columns contain zero or null:

Toggle OFF	Affected Columns	Value
ma_enabled	gdwlq, gdwlipq, aqaq	0
leasing_enabled	rouq, leaseq, xlrq	0
stock_comp_enabled	stkcpq, diluted_shares	0 / = basic_shares
workforce_detail	empq	Computed from formula, not decisio
provisions_enabled	loq (provisions component)	0

This means analysis code works on any run regardless of toggle configuration -- columns are always there, just sometimes zero.

How Toggles Affect Scoring

Scoring uses the same formulas regardless of toggles. Equity IRR is always computed from cash flows to equity investors. Debt returns are always computed from lending cash flows. The scoring system is toggle-agnostic.

The environment rating system is also toggle-agnostic -- agents rate the environment on the same 6 dimensions regardless of which features were active.

Configuration Validation

The orchestrator validates toggle combinations at run start:

Rule	Enforcement
ma_enabled requires entry_exit	Error if M&A on but entry/exit off
allow_ipo requires mode: "private_start"	Warning if IPO allowed but already public
stock_comp_enabled ignored if financial_institution	Warning (no dilution effect without market pricing)
n_firms_initial <= n_firms_max	Error
n_quarters >= 4	Error (need at least 1 year)
n_firms_initial >= 2	Error (need competition)